Felodipine ER PopPK: candidate model structures

M1–M6 equations and Pumas model code; final model M5 (fallback M3)

2026-10-25

Table of contents

Shared elements 1
M1 — 1-compartment + Weibull release (v1 baseline, 2026-10-16) 2
M2 — 2-compartment + Weibull release (2026-10-16, rejected) 3
M3 — Weibull release → first-order absorption → 2-compartment (v2, per arm) 4
M4 — M3 paired: reference and test in the same subjects (prototype) 5
M3d — M3 in the `@delay` form (2026-10-24) 6
M5 — dual Weibull release → first-order absorption → 2-compartment (final model) 7
M6 — Weibull release → 3 transit compartments → 2-compartment (tested, not selected) 8
Model selection 9

Source files: `poppk-fits/src/` (copies in `Felodipine-BCSII/poppk/src/`). The code blocks below are pasted verbatim from those files by `build_model_equations.py`.

Final model (2026-10-25): M5, dual Weibull release in the `@delay` form with the fixed release lag → first-order absorption → 2-compartment, fitted per arm on all 27 arms. **M3** stays as the fallback. The results that led to this choice are summarised in Model selection at the end.

Shared elements

Data. 100 PK-Sim subjects per arm, 18 samples each (0.5–96 h), 15 % proportional noise. Dose 10 mg = 26 024 nmol; units h, nmol, L, nmol/L.

Release lag. $t_{\text{lag}} = 0.3113$ h (fixed, PK-Sim product value); $\tau = t - t_{\text{lag}}$.

Weibull input (`@delay` format). Using the Pumas built-in delay input (`@delay(Weibull(SS, TD), Central)`), the dose enters as Dose times a Weibull density with shape SS and scale TD (Distributions.jl order `Weibull(shape, scale)`), started at the lag:

$$I(t) = \text{Dose} \cdot f_W\big(t - t_{\text{lag}}; SS, TD\big), \qquad f_W(\tau; SS, TD) = \begin{cases} \frac{SS}{TD}\left(\frac{\tau}{TD}\right)^{SS-1} \exp\left[-\left(\frac{\tau}{TD}\right)^{SS}\right], & \tau > 0 \\ 0, & \tau \le 0 \end{cases}$$

Cumulative fraction released/absorbed (`ABSO`):

$$\text{ABSO}\ (t) = 1 - \exp\left[-\left(\frac{t - t_{\text{lag}}}{TD}\right)^{SS}\right], \qquad T_{50} = TD\,(\ln 2)^{1/SS} \Leftrightarrow TD = \frac{T_{50}}{(\ln 2)^{1/SS}}$$

M1/M2 estimate the scale TD directly (code: `tvalpha`/`Alpha`); M3/M4 estimate T_{50} (code: `tvt50`), which is converted to TD. The shape SS is `tvgamma`/`Gamma` in the code.

Implementation note. Long-acting injectable models commonly use Pumas `@delay`; the felodipine code writes the same input as a depot with the Weibull hazard $h(\tau) = \frac{SS}{TD}(\tau/TD)^{SS-1}$: $\frac{dA_d}{dt} = -h(\tau)A_d$, so $A_d(t) = \text{Dose} \cdot \exp\left[-(\tau/TD)^{SS}\right]$ and $h(\tau)A_d(t) = \text{Dose} \cdot f_W(\tau) = I(t)$ exactly. In code, τ is floored at 10^{-8} h (M1, M2) or 0.01 h (M3, M4) because the density is singular at $\tau = 0$ when $SS < 1$ (clamping $SS \ge 1$ is not possible here: the felodipine arms span shapes 0.5–1.5).

`@delay` form (M3d, M5, M6). These models use Pumas `@delay` directly, as long-acting injectable models commonly do: the dose record goes into the receiving compartment with bioavailability 0 and the input is `@delay(Tlag + truncated(Weibull(SS, TD), τ0, Inf), cmt)` with $\tau_0 = 10^{-4}$ h, i.e.

$$I(t) = \text{Dose} \cdot \frac{f_W(\tau; SS, TD)}{1 - F_W(\tau_0; SS, TD)} \quad (\tau \geq \tau_0), \qquad 0 \quad \text{otherwise} .$$

Three properties of `@delay` found while building it (prototypes 2026-10-24):

- `@dosecontrol` lags are ignored by `@delay`, so the lag has to sit inside the delay distribution.
- An untruncated `Tlag + Weibull`, or a dose record placed at $t = t_{\text{lag}}$, returns an Inf likelihood for $SS < 1$ (the density is singular at the dose time).
- The truncation at τ_0 removes at most 0.4 % of the release (at $SS = 0.5$). It plays the part of the clock floor of the hazard form and is closer to the exact Weibull than the 0.01 h floor of M3.

With the lag fixed, M3d reproduces M3 on the 5 diagnostic arms (best LL within 3 units, identical Tmax and mean curves). Estimating the lag in the `@delay` form fails (Inf at the start for $SS < 1$), so t_{lag} stays fixed in all models. Run times are similar to the hazard ODE (diagnostic fits: median 34–149 s per fit vs 39–127 s).

Inter-individual variability. $P_i = \theta_P \, e^{\eta_{P,i}}$, $\boldsymbol{\eta}_i \sim \mathcal{N}(\mathbf{0}, \Omega)$, Ω diagonal.

Residual error. $y_{ij} = C_{ij}\,(1 + \varepsilon_{ij})$, $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ (implemented as SD $= \sigma C + 10^{-6}$).

Output. $C = A_c / V_c$. All volumes and clearances are apparent (/F). FOCE, two starts per fit.

M1 — 1-compartment + Weibull release (v1 baseline, 2026-10-16)

$$\frac{dA_c}{dt} = I(t) - \frac{CL}{V_c} A_c$$

$\theta = (TD, SS, CL, V_c)$; IIV on TD, CL, V_c; 8 estimated parameters (4 θ, 3 ω^2, σ).

```
poppk_model_felo = @model begin
    @param begin
        tvalpha ∈ RealDomain(lower = 0.05, init = 7.0)
        tvgamma ∈ RealDomain(lower = 0.1,  init = 0.9)
        tvcl    ∈ RealDomain(lower = 1.0,  init = 300.0)
        tvvc    ∈ RealDomain(lower = 1.0,  init = 800.0)
        tvlag   ∈ RealDomain(lower = 0.0,  init = 0.3113)
        Ω       ∈ PDiagDomain(init = [0.09, 0.09, 0.09])
        σ_p     ∈ RealDomain(lower = 0.0,  init = 0.15)
    end
    @random begin
        η ~ MvNormal(Ω)
    end
    @pre begin
        Alpha = tvalpha * exp(η[1])
        CL    = tvcl    * exp(η[2])
        Vc    = tvvc    * exp(η[3])
        Gamma = tvgamma
        Tlag  = tvlag
    end
    @dynamics begin
        Depot'   = -ifelse(t > Tlag, (Gamma/Alpha) * (max(t - Tlag, 1e-8)/Alpha)^(Gamma
- 1), 0.0) * Depot
        Central' =  ifelse(t > Tlag, (Gamma/Alpha) * (max(t - Tlag, 1e-8)/Alpha)^(Gamma
- 1), 0.0) * Depot - (CL / Vc) * Central
    end
```

```
    @derived begin
        cp := @. Central / Vc
        dv ~ @. Normal(cp, abs(cp) * σ_p + 1e-6)
    end
end

# The two starting points (see header).
const INITS_FELO = [
    "release_limited"     => (tvalpha = 13.0, tvgamma = 1.0, tvcl = 300.0, tvvc =
250.0),
    "disposition_limited" => (tvalpha = 1.0,  tvgamma = 1.3, tvcl = 300.0, tvvc =
4000.0)]
```

M2 — 2-compartment + Weibull release (2026-10-16, rejected)

$$\frac{dA_c}{dt} = I(t) - \frac{CL + Q}{V_c} A_c + \frac{Q}{V_p} A_p$$

$$\frac{dA_p}{dt} = \frac{Q}{V_c} A_c - \frac{Q}{V_p} A_p$$

$\theta = (TD, SS, CL, V_c, Q, V_p)$; IIV on TD, CL, V_c; 10 estimated parameters. On 2026-10-16, Q or V_p ran to their bounds from either start (peripheral compartment not identified).

```
poppk_model_felo = @model begin
    @param begin
        tvalpha ∈ RealDomain(lower = 0.05, init = 7.0)    # Weibull release scale (h)
        tvgamma ∈ RealDomain(lower = 0.1,  init = 0.9)    # Weibull shape
        tvcl    ∈ RealDomain(lower = 1.0,  init = 300.0)  # CL/F (L/h)
        tvvc    ∈ RealDomain(lower = 1.0,  init = 800.0)  # Vc/F (L)
        tvq     ∈ RealDomain(lower = 1.0,  init = 400.0)  # Q/F (L/h)
        tvvp    ∈ RealDomain(lower = 1.0,  init = 4000.0) # Vp/F (L)
        tvlag   ∈ RealDomain(lower = 0.0,  init = 0.3113) # release lag (h), held
constant
        Ω       ∈ PDiagDomain(init = [0.09, 0.09, 0.09])  # IIV: Alpha, CL, Vc
        σ_p     ∈ RealDomain(lower = 0.0,  init = 0.15)   # proportional residual SD
    end
    @random begin
        η ~ MvNormal(Ω)
    end
    @pre begin
        Alpha = tvalpha * exp(η[1])
        CL    = tvcl    * exp(η[2])
        Vc    = tvvc    * exp(η[3])
        Q     = tvq
        Vp    = tvvp
        Gamma = tvgamma
        Tlag  = tvlag
    end
    @dynamics begin
        Depot'   = -ifelse(t > Tlag, (Gamma/Alpha) * (max(t - Tlag, 1e-8)/Alpha)^(Gamma
- 1), 0.0) * Depot
        Central' =  ifelse(t > Tlag, (Gamma/Alpha) * (max(t - Tlag, 1e-8)/Alpha)^(Gamma
- 1), 0.0) * Depot -
                    (CL / Vc) * Central - (Q / Vc) * Central + (Q / Vp) * Periph
        Periph'  =  (Q / Vc) * Central - (Q / Vp) * Periph
    end
```

```
    @derived begin
        cp := @. Central / Vc
        dv ~ @. Normal(cp, abs(cp) * σ_p + 1e-6)
    end
end
```

M3 — Weibull release → first-order absorption → 2-compartment (v2, per arm)

$$\frac{dA_g}{dt} = I(t) - k_a\, A_g$$
$$\frac{dA_c}{dt} = k_a\, A_g - \frac{CL + Q}{V_c}\, A_c + \frac{Q}{V_p}\, A_p$$
$$\frac{dA_p}{dt} = \frac{Q}{V_c}\, A_c - \frac{Q}{V_p}\, A_p$$

$$T_{50,i} = \theta_{T_{50}}\, e^{\eta_{T_{50},i}}, \qquad CL_i = \theta_{CL}\, e^{\eta_{CL,i}}, \qquad V_{c,i} = \theta_{V_c}\, e^{\eta_{V_c,i}}$$

$\theta = \left(T_{50}, SS, k_a, CL, V_c, Q, V_p\right)$; 11 estimated parameters (7 θ, 3 ω^2, σ). Starts: T_{50} and SS at the arm's PK-Sim values, plus the two disposition starts below. The gut compartment separates release from appearance in plasma, giving the PBPK profile's fast rise and plateau without forcing release to be fast.

```
poppk_felo_wka2 = @model begin
    @param begin
        tvt50   ∈ RealDomain(lower = 0.05, init = 4.0)
        tvgamma ∈ RealDomain(lower = 0.1,  init = 0.8)
        tvka    ∈ RealDomain(lower = 0.01, init = 1.0)
        tvcl    ∈ RealDomain(lower = 1.0,  init = 300.0)
        tvvc    ∈ RealDomain(lower = 1.0,  init = 200.0)
        tvq     ∈ RealDomain(lower = 0.1,  init = 150.0)
        tvvp    ∈ RealDomain(lower = 1.0,  init = 1000.0)
        tvlag   ∈ RealDomain(lower = 0.0,  init = 0.3113)
        Ω       ∈ PDiagDomain(init = [0.05, 0.09, 0.09])
        σ_p     ∈ RealDomain(lower = 0.0,  init = 0.15)
    end
    @random begin
        η ~ MvNormal(Ω)
    end
    @pre begin
        T50 = tvt50 * exp(η[1]); CL = tvcl * exp(η[2]); Vc = tvvc * exp(η[3])
        Gamma = tvgamma; Ka = tvka; Q = tvq; Vp = tvvp; Tlag = tvlag
        Alpha = T50 / log(2)^(1 / Gamma)          # Weibull scale from T50
    end
    @dynamics begin
        Depot'   = -ifelse(t > Tlag, (Gamma/Alpha) * (max(t - Tlag, 0.01)/Alpha)^(Gamma
- 1), 0.0) * Depot
        Gut'     =  ifelse(t > Tlag, (Gamma/Alpha) * (max(t - Tlag, 0.01)/Alpha)^(Gamma
- 1), 0.0) * Depot - Ka * Gut
        Central' =  Ka * Gut - (CL + Q) / Vc * Central + Q / Vp * Periph
        Periph'  =  Q / Vc * Central - Q / Vp * Periph
    end
    @derived begin
        cp := @. Central / Vc
        dv ~ @. Normal(cp, abs(cp) * σ_p + 1e-6)
```

```
    end
end

const INITS_V2 = [
    "release_limited"    => (tvka = 2.0, tvcl = 300.0, tvvc = 200.0, tvq = 200.0, tvvp
= 1500.0),
    "absorption_limited" => (tvka = 0.3, tvcl = 300.0, tvvc = 500.0, tvq = 100.0, tvvp
= 1000.0)]
```

M4 — M3 paired: reference and test in the same subjects (prototype)

Reference dosed at $t = 0$, test at $t = 200$ h, each through its own Weibull input: $I_R(t) = \text{Dose} \cdot f_W\left(t - t_{\text{lag}}; SS_R, TD_R\right)$ and $I_T(t) = \text{Dose} \cdot f_W\left(t - 200 - t_{\text{lag}}; SS_T, TD_T\right)$ (code: `Depot1`, `Depot2`).

$$\frac{dA_g}{dt} = I_R(t) + I_T(t) - k_a\, A_g$$
$$\frac{dA_c}{dt} = k_a\, A_g - \frac{CL + Q}{V_c}\, A_c + \frac{Q}{V_p}\, A_p$$
$$\frac{dA_p}{dt} = \frac{Q}{V_c}\, A_c - \frac{Q}{V_p}\, A_p$$

$TD_R = T_{50,R}/(\ln 2)^{1/SS_R}$, $TD_T = T_{50,T}/(\ln 2)^{1/SS_T}$. One subject-level $\eta_{T_{50}}$ scales both $T_{50,R}$ and $T_{50,T}$; k_a, CL, V_c, Q, V_p and η_{CL}, η_{V_c} are shared across periods. 13 estimated parameters. Its AIC is not comparable with the per-arm models (two arms of data); compare its LL with the sum of the M3 reference-arm and test-arm LLs.

```
const TP2 = 200.0   # period-2 dose time in the paired dataset (washout >> 5 t½)

poppk_felo_pair = @model begin
    @param begin
        t50R    ∈ RealDomain(lower = 0.05, init = 4.0)
        gR      ∈ RealDomain(lower = 0.1,  init = 0.8)
        t50T    ∈ RealDomain(lower = 0.05, init = 4.0)
        gT      ∈ RealDomain(lower = 0.1,  init = 0.8)
        tvka    ∈ RealDomain(lower = 0.01, init = 1.0)
        tvcl    ∈ RealDomain(lower = 1.0,  init = 300.0)
        tvvc    ∈ RealDomain(lower = 1.0,  init = 200.0)
        tvq     ∈ RealDomain(lower = 0.1,  init = 150.0)
        tvvp    ∈ RealDomain(lower = 1.0,  init = 1000.0)
        tvlag   ∈ RealDomain(lower = 0.0,  init = 0.3113)
        Ω       ∈ PDiagDomain(init = [0.05, 0.09, 0.09])
        σ_p     ∈ RealDomain(lower = 0.0,  init = 0.15)
    end
    @random begin
        η ~ MvNormal(Ω)
    end
    @pre begin
        T50R = t50R * exp(η[1]); T50T = t50T * exp(η[1])   # subject release eta shared
(GI transit)
        CL = tvcl * exp(η[2]); Vc = tvvc * exp(η[3])
        Ka = tvka; Q = tvq; Vp = tvvp; Tlag = tvlag; GR = gR; GT = gT
        AR = T50R / log(2)^(1 / GR); AT = T50T / log(2)^(1 / GT); Tp2 = TP2
    end
    @dynamics begin
        Depot1'  = -ifelse(t > Tlag, (GR/AR) * (max(t - Tlag, 0.01)/AR)^(GR - 1), 0.0)
```

```
* Depot1
        Depot2'  = -ifelse(t > Tp2 + Tlag, (GT/AT) * (max(t - Tp2 - Tlag, 0.01)/AT)^(GT
- 1), 0.0) * Depot2
        Gut'     =  ifelse(t > Tlag, (GR/AR) * (max(t - Tlag, 0.01)/AR)^(GR - 1), 0.0)
* Depot1 +
                    ifelse(t > Tp2 + Tlag, (GT/AT) * (max(t - Tp2 - Tlag, 0.01)/AT)^(GT
- 1), 0.0) * Depot2 - Ka * Gut
        Central' =  Ka * Gut - (CL + Q) / Vc * Central + Q / Vp * Periph
        Periph'  =  Q / Vc * Central - Q / Vp * Periph
    end
    @derived begin
        cp := @. Central / Vc
        dv ~ @. Normal(cp, abs(cp) * σ_p + 1e-6)
    end
end
```

M3d — M3 in the `@delay` form (2026-10-24)

The same equations as M3, with $I(t)$ written as the truncated `@delay` input above (dose record into `Gut`, bioavailability 0). 11 estimated parameters, the M3 starts. Used to check that the delay-style input gives the same fit as the hazard ODE before building M5 and M6 in that form.

```
poppk_m3d = @model begin
    @param begin
        tvt50   ∈ RealDomain(lower = 0.05, init = 4.0)
        tvgamma ∈ RealDomain(lower = 0.1,  init = 0.8)
        tvka    ∈ RealDomain(lower = 0.01, init = 1.0)
        tvcl    ∈ RealDomain(lower = 1.0,  init = 300.0)
        tvvc    ∈ RealDomain(lower = 1.0,  init = 200.0)
        tvq     ∈ RealDomain(lower = 0.1,  init = 150.0)
        tvvp    ∈ RealDomain(lower = 1.0,  init = 1000.0)
        Ω       ∈ PDiagDomain(init = [0.05, 0.09, 0.09])
        tvlag   ∈ RealDomain(lower = 0.0,  init = 0.3113)
        σ_p     ∈ RealDomain(lower = 0.0,  init = 0.15)
    end
    @random begin
        η ~ MvNormal(Ω)
    end
    @pre begin
        T50 = tvt50 * exp(η[1]); CL = tvcl * exp(η[2]); Vc = tvvc * exp(η[3])
        Gamma = tvgamma; Ka = tvka; Q = tvq; Vp = tvvp
        Alpha = T50 / log(2)^(1 / Gamma)
        Tlag  = tvlag; Tau0 = 1e-4                # truncation, see header
        dd    = @delay(Tlag + truncated(Weibull(Gamma, Alpha), Tau0, Inf), Gut)
    end
    @dosecontrol begin
        bioav = (; Gut = 0.0,)
    end
    @dynamics begin
        Gut'     = dd - Ka * Gut
        Central' = Ka * Gut - (CL + Q) / Vc * Central + Q / Vp * Periph
        Periph'  = Q / Vc * Central - Q / Vp * Periph
    end
    @derived begin
        cp := @. Central / Vc
        dv ~ @. Normal(cp, abs(cp) * σ_p + 1e-6)
```

```
    end
end
```

M5 — dual Weibull release → first-order absorption → 2-compartment (final model)

M3 peaks 0.4–0.7 h early on most arms and falls too fast at high shape: one release timescale and one k_a cannot give both the fast initial phase and the prolonged late release of the PBPK profile. M5 replaces the single Weibull with a two-Weibull mixture (`FF*dd1 + (1-FF)*dd2`):

$$I(t) = \text{Dose}\,\left[FF\cdot \tilde{f}_W(\tau; SS_a, TD_a) + (1-FF)\cdot \tilde{f}_W(\tau; SS_b, TD_b)\right], \qquad TD_k = \frac{T_{50,k}}{(\ln 2)^{1/SS_k}}$$

where $\tilde{f}_W$ is the Weibull density truncated at τ_0 (see the `@delay` form above), and

$$\begin{aligned}
\frac{dA_g}{dt} &= I(t) - k_a\,A_g \\
\frac{dA_c}{dt} &= k_a\,A_g - \frac{CL+Q}{V_c}\,A_c + \frac{Q}{V_p}\,A_p \\
\frac{dA_p}{dt} &= \frac{Q}{V_c}\,A_c - \frac{Q}{V_p}\,A_p
\end{aligned}$$

$$T_{50a,i} = \theta_{T_{50a}}\,e^{\eta_{T_{50},i}}, \quad T_{50b,i} = \theta_{T_{50b}}\,e^{\eta_{T_{50},i}}, \quad CL_i = \theta_{CL}\,e^{\eta_{CL,i}}, \quad V_{c,i} = \theta_{V_c}\,e^{\eta_{V_c,i}}$$

$\theta = \left(T_{50a}, SS_a, T_{50b}, SS_b, FF, k_a, CL, V_c, Q, V_p\right)$, $0 \le FF \le 1$; one release η scales both components; $t_{\text{lag}} = 0.3113$ h fixed. 14 estimated parameters (10 θ, 3 ω^2, σ).

Starts. Two starts bracket the arm's PK-Sim T_{50} (T) and shape (s). The 27-arm production fits (`02c_fit_poppk_m5.jl`) use the absorption-limited start only: on the 7 arms fitted from both starts it was never worse than the release-limited one and had the higher LL on 3. Fits take about 30 min per arm single-threaded.

```
starts(t, s) = [
    "release_limited"    => (tvt50a = 0.5t, tvga = s, tvt50b = 3t, tvgb = s, tvff =
0.6, tvka = 2.0,
                             tvcl = 300.0, tvvc = 200.0, tvq = 200.0, tvvp = 1500.0,
tvlag = 0.3113),
    "absorption_limited" => (tvt50a = 0.3t, tvga = 1.0, tvt50b = 2t, tvgb = 1.0, tvff =
0.3, tvka = 0.5,
                             tvcl = 300.0, tvvc = 500.0, tvq = 100.0, tvvp = 1000.0,
tvlag = 0.3113)]
```

Interpretation. The two components are a flexible description of the release curve, not two physical populations: FF ranges 0.02–0.97 across arms, and on some arms component a is a small fast burst (T_{50a} 0.2–0.4 h). For comparison with the PK-Sim input, the per-arm parameter table reports summaries of the mixture: `tvt50` = time of 50 % released (lag excluded) and `tvgamma` = shape of the single Weibull through the mixture's 25 % and 75 % release times. Like M3's T_{50}, the summary T_{50} sits well above the PK-Sim dissolution T_{50} (≈ 8–9 h at PK-Sim 4–5 h) because the PopPK input lumps release, gut transit and absorption. At the reference arm FF (RSE 45 %) and Q, V_p (RSE 55–70 %) are poorly determined; V_c reaches ≈ 2300 L at shape 1.5, as in M3.

```
poppk_m5 = @model begin
    @param begin
```

```
        tvt50a  ∈ RealDomain(lower = 0.05, init = 2.0)     # fast release component
        tvga    ∈ RealDomain(lower = 0.1,  init = 0.8)
        tvt50b  ∈ RealDomain(lower = 0.05, init = 10.0)    # slow release component
        tvgb    ∈ RealDomain(lower = 0.1,  init = 0.8)
        tvff    ∈ RealDomain(lower = 0.0, upper = 1.0, init = 0.5)   # fraction in the
fast component
        tvka    ∈ RealDomain(lower = 0.01, init = 1.0)
        tvcl    ∈ RealDomain(lower = 1.0,  init = 300.0)
        tvvc    ∈ RealDomain(lower = 1.0,  init = 200.0)
        tvq     ∈ RealDomain(lower = 0.1,  init = 150.0)
        tvvp    ∈ RealDomain(lower = 1.0,  init = 1000.0)
        Ω       ∈ PDiagDomain(init = [0.05, 0.09, 0.09])
        tvlag   ∈ RealDomain(lower = 0.0,  init = 0.3113)
        σ_p     ∈ RealDomain(lower = 0.0,  init = 0.15)
    end
    @random begin
        η ~ MvNormal(Ω)
    end
    @pre begin
        T50a = tvt50a * exp(η[1]); T50b = tvt50b * exp(η[1])
        CL = tvcl * exp(η[2]); Vc = tvvc * exp(η[3])
        Ga = tvga; Gb = tvgb; FF = tvff; Ka = tvka; Q = tvq; Vp = tvvp
        Aa = T50a / log(2)^(1 / Ga); Ab = T50b / log(2)^(1 / Gb)
        Tlag = tvlag; Tau0 = 1e-4                # truncation, see
poppk_models_felo_delay.jl
        dd1  = @delay(Tlag + truncated(Weibull(Ga, Aa), Tau0, Inf), Gut)
        dd2  = @delay(Tlag + truncated(Weibull(Gb, Ab), Tau0, Inf), Gut)
    end
    @dosecontrol begin
        bioav = (; Gut = 0.0,)
    end
    @dynamics begin
        Gut'     = FF * dd1 + (1 - FF) * dd2 - Ka * Gut
        Central' = Ka * Gut - (CL + Q) / Vc * Central + Q / Vp * Periph
        Periph'  = Q / Vc * Central - Q / Vp * Periph
    end
    @derived begin
        cp := @. Central / Vc
        dv ~ @. Normal(cp, abs(cp) * σ_p + 1e-6)
    end
end
```

M6 — Weibull release → 3 transit compartments → 2-compartment (tested, not selected)

$$\frac{dA_1}{dt} = I(t) - k_{tr}\,A_1, \qquad \frac{dA_k}{dt} = k_{tr}\,(A_{k-1} - A_k), k = 2, 3$$

$$\frac{dA_c}{dt} = k_{tr}\,A_3 - \frac{CL + Q}{V_c}\,A_c + \frac{Q}{V_p}\,A_p, \qquad \frac{dA_p}{dt} = \frac{Q}{V_c}\,A_c - \frac{Q}{V_p}\,A_p$$

$I(t)$ as M3d; the transit chain replaces k_a. 11 estimated parameters. It fixes the shape-1.5 curve (LL 19 vs M3 −59) but is no better than M3 on the other diagnostic arms, and 2 of its 10 fits failed.

```
poppk_m6 = @model begin
    @param begin
```

```
        tvt50   ∈ RealDomain(lower = 0.05, init = 4.0)
        tvgamma ∈ RealDomain(lower = 0.1,  init = 0.8)
        tvktr   ∈ RealDomain(lower = 0.01, init = 3.0)     # transit rate (3
compartments)
        tvcl    ∈ RealDomain(lower = 1.0,  init = 300.0)
        tvvc    ∈ RealDomain(lower = 1.0,  init = 200.0)
        tvq     ∈ RealDomain(lower = 0.1,  init = 150.0)
        tvvp    ∈ RealDomain(lower = 1.0,  init = 1000.0)
        Ω       ∈ PDiagDomain(init = [0.05, 0.09, 0.09])
        tvlag   ∈ RealDomain(lower = 0.0,  init = 0.3113)
        σ_p     ∈ RealDomain(lower = 0.0,  init = 0.15)
    end
    @random begin
        η ~ MvNormal(Ω)
    end
    @pre begin
        T50 = tvt50 * exp(η[1]); CL = tvcl * exp(η[2]); Vc = tvvc * exp(η[3])
        Gamma = tvgamma; Ktr = tvktr; Q = tvq; Vp = tvvp
        Alpha = T50 / log(2)^(1 / Gamma)
        Tlag  = tvlag; Tau0 = 1e-4
        dd    = @delay(Tlag + truncated(Weibull(Gamma, Alpha), Tau0, Inf), Tr1)
    end
    @dosecontrol begin
        bioav = (; Tr1 = 0.0,)
    end
    @dynamics begin
        Tr1'     = dd - Ktr * Tr1
        Tr2'     = Ktr * Tr1 - Ktr * Tr2
        Tr3'     = Ktr * Tr2 - Ktr * Tr3
        Central' = Ktr * Tr3 - (CL + Q) / Vc * Central + Q / Vp * Periph
        Periph'  = Q / Vc * Central - Q / Vp * Periph
    end
    @derived begin
        cp := @. Central / Vc
        dv ~ @. Normal(cp, abs(cp) * σ_p + 1e-6)
    end
end
```

Model selection

Diagnostic arms (4.78/0.764 reference, 4.78/0.5, 4.78/1.5, 2.39/0.764, 9.56/0.764; best start per arm, mean curves and per-subject Tmax against the noise-free PK-Sim profiles):

Model	Best LL by arm (ref, 0.5, 1.5, 2.39, 9.56)	Tmax bias (median per subject)	Mean-curve error 0.5–12 h
M3	−354, −487, −59, −481, −108	−0.7 to +0.1 h	2.0–6.6 %
M3d	−354, −488, −56, −481, −107	as M3	as M3
M3, no lag (`@delay`)	−755, −754, −48, −866, −500	+60 % overshoot at 0.5 h	2.9–8.4 %
M3, lag estimated (ODE)	−526, −762, −329, −808, −106	lag ≈ 0.27–0.33 h, LL worse	—

Model	Best LL by arm (ref, 0.5, 1.5, 2.39, 9.56)	Tmax bias (median per subject)	Mean-curve error 0.5–12 h
M5	**−348, −439, +38, −433, −85**	**−0.3 to +0.15 h**	**1.2–4.6 %**
M6	−365, −479, +19, −484, −109	−0.45 to 0 h	2.1–4.3 %

All 27 arms, M5 vs M3 (`felo_09_m5_vs_m3_all27.jl`; Cmax/AUC against the PBPK values in the same subjects, T/R GMR against the 4.78/0.764 reference arm):

	M3 (fallback)	M5 (final)
AIC better	0 / 27 arms	27 / 27 (median ΔAIC −48, range −289 to −6.5)
Cmax ratio PopPK/PBPK	0.976–1.043	0.976–1.019
AUC_{0-t} ratio PopPK/PBPK	0.985–1.012	0.989–1.004
Cmax T/R GMR error vs PBPK	max 3.5 %, mean 1.6 %	max 3.3 %, mean 1.3 %
Tmax bias vs PK-Sim (median of arms)	0.45 h	0.20 h
Subjects with Tmax within 0.5 h	64 %	76 %
2-D safe-space T50 limits vs PBPK	mean |Δ| 0.08 h, max 0.20 h, bias +0.03 h	mean |Δ| 0.09 h, max 0.25 h, bias −0.09 h

M5 describes each arm better, but its safe-space limits are not closer to PBPK than M3′s. M5 overpredicts the reference-arm Cmax by 0.9 % while its test arms are nearly unbiased, so its T/R GMRs run about 1.5 % low at shape ≥ 0.9 and the limits shift left. M3 overpredicts Cmax on the reference and test arms alike (up to 3.5 % at shape 1.2) and the two largely cancel. Either model leaves the BE conclusions unchanged (all limits within 0.25 h of PBPK). One M5 arm (6.31/1.2) still peaks early (median −0.85 h).

BE decision vs PBPK (`felo_04_pbpk_poppk_compare.jl`; 1-D axes through the reference, WSV fasted, Type I from 20 000 simulated trials, seed 1234). Cmax sets every limit; AUC_{0-t} never reaches the BE bounds.

	T50 axis lower limit (h)	T50 axis upper limit (h)	Type I at declared limit, Cmax, n = 24 / 54 (lower; upper)
PBPK	3.25	6.96	4.9 / 5.1 %; 4.7 / 5.0 % (nominal)
M3	3.40	6.83	7.6 / 10.2 %; 5.8 / 6.8 %
M5	3.27	6.86	5.3 / 5.8 %; 5.6 / 6.3 %

On the shape axis (T50 4.78 h) PBPK loses BE at shape 1.465. Both PopPK models underpredict the shape-1.5 Cmax GMR (M3 1.235, M5 1.227, PBPK 1.269), so neither reaches 1.25 within the tested range.